

Tutorial 01 – Instructor notes

ENVX2001 – Applied Statistical Methods

Semester 1

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Companion document for tutors running Tutorial 01. Mirrors each exercise with talking points, expected output at each checkpoint, and common problems.

Student document: [Tutorial 01](#)

Before you start

Get these sorted before students arrive:

- Have RStudio open with a fresh session.
- Download all four CSV files from the student tutorial page. You only need one for your demo, but students will pick different ones and ask about theirs.
- Decide which dataset you'll demo with. Ghibli is the default in the student doc, so it's easiest to follow along with.
- Skim the student document if you haven't recently.

Exercise 1: Set up your project

What to do

Demo creating an RStudio Project: **File > New Project... > New Directory > New Project**. Name it tutorial01-demo or similar. Create a data/ folder in the Files pane and move your CSV into it.

What to say

- RStudio Projects set the working directory automatically, so data/myfile.csv works without a full path.
- This is how they should set up every lab and assignment — one project per task.
- Show where the .Rproj file lives and what it does (sets the working directory, nothing more).

Expected output

The Files pane shows the project folder with a data/ subfolder containing one CSV file.

Common issues

Warning

CSV ends up in Downloads instead of data/. Students download the file but forget to move it. Show them how to drag the file from their Downloads folder into the data/ folder. On Windows, the “More > Set As Working Directory” option in the Files pane sometimes confuses things — stick to the system file manager.

Warning

Data instead of data. Case matters on macOS and Linux. Not a problem on Windows, but good to keep it lowercase from the start.

Exercise 2: Create your document

What to do

Demo File > New File > Quarto Document... > Create Empty Document. Paste in the YAML header, save as `report.qmd`, and render with `Cmd+Shift+K` (Mac) or `Ctrl+Shift+K` (Windows/Linux).

```
CODE
---
title: "Exploring My Dataset"
author: "Your Name"
date: today
format:
  html:
    theme: cosmo
    toc: true
---
```

What to say

- Walk through each YAML field: `title` and `author` are plain text, `date: today` auto-fills the current date, `format` controls the output.
- **Indentation matters** in YAML. `html:` must be indented under `format:`, and `theme:` under `html:`. Two spaces per level, no tabs.
- Render the empty doc first. This shows students the feedback loop early: edit, save, render, check.

Expected output

An HTML page with the title, author name, today's date, and an empty body. The table of contents sidebar is visible but empty.

Common issues

Warning

YAML indentation errors. The most common problem in this exercise. Students copy-paste the YAML and indentation gets mangled. Tell them to delete and retype the spaces if rendering fails.

Warning

“Quarto Document” option missing. Student has an old RStudio version or Quarto isn’t installed. They need RStudio 2022.07 or later. As a workaround, they can create a plain text file and save it as `.qmd`.

Exercise 3: Write about your data

What to do

Demo adding markdown text below the YAML. Type a heading, some bold text, a bullet list. Use the example from the student doc or make up your own.

What to say

- Show the syntax live: `##` for headings, `**bold**`, `*italic*`, `-` for bullets.
- Let students write their own content. Don’t dictate what they write — this is their chance to practise markdown without worrying about code.
- Remind them to leave a blank line before headings and between paragraphs. Quarto won’t render a heading properly if there’s no blank line above it.
- Render after writing to show the table of contents populating.

Expected output

Formatted text with headings, bold, italic, and a bullet list. The table of contents sidebar lists the new sections.

Common issues

Warning

No blank line before a heading. The heading renders as plain text or merges with the paragraph above. Show what this looks like so students can recognise it next time.

⚠ Warning

Students unsure what to write. Tell them to just describe their dataset: what's in it, what looks interesting, what surprised them. Doesn't need to be long.

Exercise 4: Load and display your data

What to do

Demo adding the setup chunk first (`library(tidyverse)` with `message: false`), then a data-loading chunk. Render and show the output. Pick whichever dataset you like for your demo – the tabs below have the code for all four, but you only need one.

Code for each dataset

Ghibli

```
CODE
library(tidyverse)
my_data <- read_csv("data/ghibli_films.csv")
my_data
```

Columns: title, year, rating, runtime

Spotify

```
CODE
library(tidyverse)
my_data <- read_csv("data/spotify_top50.csv")
my_data
```

Columns: title, artist, streams_millions, genre

AFL

```
CODE
library(tidyverse)
my_data <- read_csv("data/afl_teams.csv")
my_data
```

Columns: team, wins_2024, losses_2024, premierships, founded

Fast food

CODE

```
library(tidyverse)
my_data <- read_csv("data/fast_food.csv")
my_data
```

Columns: name, rating, price_range, locations_au

What to say

- `library(tidyverse)` loads the packages we need. `read_csv()` comes from `readr`, which is part of the tidyverse.
- The `message: false` and `warning: false` chunk options hide startup messages. Show what happens without them so students understand why they're useful.
- `read_csv()` uses a relative path from the project root. That's why setting up the project first matters.

Expected output

The setup chunk produces no visible output (messages suppressed). The data chunk prints a tibble showing all rows and columns.

Common issues

⚠ Warning

"could not find function 'read_csv'" Student forgot to run the setup chunk or didn't include `library(tidyverse)`. Show the error so they learn to recognise it.

⚠ Warning

"file does not exist" Wrong file name (capitalisation, typo) or the CSV isn't in `data/`. Have them check the Files pane — the file name there is exactly what they need to type.

⚠ Warning

`read.csv()` instead of `read_csv()` Works fine, but returns a `data.frame` instead of a tibble. Not worth correcting in week 1 — just mention that `read_csv()` is the tidyverse version and we'll use it through the semester.

Exercise 5: Create a chart

What to do

Demo building a ggplot bar chart. Type it line by line so students can follow along. Render and show the result. Use the same dataset you picked for Exercise 4 – the other variants are here for reference if students ask.

Code for each dataset

Ghibli

CODE

```
ggplot(my_data, aes(x = reorder(title, rating), y = rating)) +  
  geom_col() +  
  coord_flip() +  
  labs(x = NULL, y = "Rating")
```

Spotify

CODE

```
ggplot(my_data, aes(x = reorder(title, streams_millions), y = streams_millions)) +  
  geom_col() +  
  coord_flip() +  
  labs(x = NULL, y = "Streams (millions)")
```

AFL

CODE

```
ggplot(my_data, aes(x = reorder(team, premierships), y = premierships)) +  
  geom_col() +  
  coord_flip() +  
  labs(x = NULL, y = "Premierships")
```

Fast food

CODE

```
ggplot(my_data, aes(x = reorder(name, rating), y = rating)) +  
  geom_col() +  
  coord_flip() +  
  labs(x = NULL, y = "Rating")
```

What to say

- Walk through each line: `aes()` maps columns to axes, `reorder()` sorts the bars, `geom_col()` draws them, `coord_flip()` flips to horizontal, `labs()` sets axis labels.
- Without `reorder()`, bars appear in alphabetical order. Show this if you have time.
- Students using non-Ghibli datasets need to swap column names. Point them to the margin note in the student doc, or to the tabset above.

Expected output

A horizontal bar chart sorted from lowest to highest, with a caption underneath. Quarto auto-numbers the figure because of the `fig-chart` label.

Common issues

⚠ Warning

Wrong column names. Students using Spotify try `rating` instead of `streams_millions`, or AFL students use `name` instead of `team`. The error says “object not found”. Tell them to check their column names from the data output in Exercise 4.

⚠ Warning

Forgot to update `labs()`. The chart renders fine but the y-axis says “Rating” when it should say “Streams (millions)” or “Premierships”. Not a breaking error, but worth catching.

⚠ Warning

Bars are unsorted. Student wrote `title` instead of `reorder(title, rating)`. The bars appear alphabetically.

Exercise 6: Add a formatted table

What to do

Demo adding a `knitr::kable()` chunk with `echo: false`. Render and show the clean table. Then go back to the chart chunk from Exercise 5 and add `echo: false` there too — render again to show the difference.

```
CODE
knitr::kable(my_data)
```

This code works the same for all four datasets.

What to say

- `echo: false` hides the source code from the rendered output. The reader sees the table but not the R code.
- This is how you'd present results in a report or assignment — code hidden, output clean.
- `knitr::kable()` produces a simple formatted table. Fancier options exist (`gt`, `flextable`), but `kable` is enough for now.
- Show `echo: true` (the default) alongside `echo: false` if you want to make the difference obvious.

Expected output

A formatted table with a caption and no visible code. If the chart chunk also has `echo: false`, the document reads like a finished report.

Common issues

⚠ Warning

Chunk option typo. `echo: False` (capital F) or `echo = false` (equals sign instead of colon). Quarto chunk options use `#| key: value` syntax — lowercase, colon, space.

Putting it all together

What to do

This section is tutor-led. Demo switching the output format live:

1. Change the `format` section of the YAML to Word output:

```
CODE
format:
  docx:
    toc: true
```

Render and show the Word document.

2. Change it to Typst (PDF) output:

```
CODE
format:
  typst:
    toc: true
```

Render and show the PDF.

3. Switch back to HTML.

What to say

- Same content, different format. Only the YAML changes.
- Typst is Quarto's newer PDF engine. It doesn't need a LaTeX installation, which saves a lot of setup pain.
- Word output is handy for collaborating with people who don't use R or Quarto.
- Students can list multiple formats in the YAML, but that's a topic for later weeks.

Expected output

The same document rendered as HTML, Word, and PDF. Content is identical; only styling differs.

Common issues

Warning

YAML indentation when switching formats. Students who copy-paste the format block can end up with mixed indentation. Same fix as before: two spaces, no tabs, `toc:` indented under the format name.